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Beauty clinic patient diagnosis prediction based on Retrieval-Augmented Generation and Large Language Models

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Abstract

Indonesia's beauty clinic sector faces challenges in cost transparency and diagnosis accuracy. This study proposes an AI solution using Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) to enhance patient diagnosis prediction. The model integrates data retrieval with natural language processing for precise predictions and transparent cost estimations. Evaluation on 20 simulated queries shows promising results: Semantic Similarity (BERTScore F1) of 0.73, Service Precision of 0.73, Service Recall of 0.72, Service F1 Score of 0.72, and Cost Accuracy of 0.72. This technology aims to boost service efficiency, increase patient satisfaction,

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Keywords: Beauty Diagnosis; RAG; LLMs; LangChain; Beauty Prediction

1. Introduction

The beauty clinic sector in Indonesia is growing rapidly, yet faces significant hurdles in treatment cost transparency and initial diagnosis accuracy [1, 2]. This lack of clarity can confuse potential patients and undermine marketing efforts, especially in a competitive market where effective promotion is crucial [3]. Patients often hesitate due to unclear cost estimations, despite various appealing offers. Therefore, improving communication through clear and transparent cost estimations is vital for building trust and increasing market share [4].

Artificial Intelligence (AI)-based prediction models offer a promising solution. This study proposes using Retrieval-Augmented Generation (RAG) LangChain combined with Large Language Models (LLMs). This approach integrates data retrieval with natural language processing to generate accurate diagnostic predictions and efficient treatment cost estimations. By leveraging this model, beauty clinics can predict diagnoses from patient data, providing precise cost estimates and explanations.

Recent research highlights the effectiveness of RAG and LLMs in various domains. Hussien et al. utilized RAG with Knowledge Graphs and LLMs for road user behavior prediction, demonstrating improved F1 scores and accuracy

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through contextual integration and Bayesian inference [4]. Gu et al. developed a generation-augmented retrieval (GAR) and Hierarchical Chain-of-Thought (HCoT) method to enhance LLMs for automated clinical assessment using Electronic Health Records (EHR). Their GAPrompt paradigm significantly improved stroke severity assessment accuracy and reduced hallucinations, demonstrating LLM optimization for EHR analysis [5]. This growing body of work underscores the potential of RAG-LLM systems to respond quickly to market demands and optimize strategies in competitive industries.

2. Related Work

Research on AI applications spans various fields, from autonomous driving to healthcare and beauty. Hussien et al. utilized a Retrieval-Augmented Generation (RAG) approach with Knowledge Graphs and Large Language Models (LLMs) to improve road user behavior prediction, showing increased F1 scores and accuracy [4]. In education, Alsafari et al. demonstrated that RAG-based chatbots with LangChain and GPT-3.5 Turbo were more adaptive and relevant for teaching assistants than traditional intent-based approaches [6].

In the medical domain, Gu et al. developed a generation-augmented retrieval (GAR) and Hierarchical Chain-of-Thought (HCoT) method to enhance LLMs for automated clinical assessment using Electronic Health Records (EHR), which significantly improved stroke severity assessment accuracy and reduced hallucinations [5]. Their work also focused on optimizing LLMs for EHR analysis, specifically for automatic stroke severity assessment, leading to improved accuracy. Further advancements in LLMs include DeepSeek-R1, an AI model that exhibits human-like reasoning and superior performance in chemistry, mathematics, and programming, rivaling OpenAI models [7]. Additionally, Li et al.'s comparative study of GPT-4o and LLaMA 3-70b indicated comparable diagnostic accuracy for radiology cases, suggesting the potential of open-source LLMs in medical applications and emphasizing the importance of model validation in clinical practice [8]. Beyond general medical applications, LLMs have also shown promise in specialized healthcare areas, such as predictive medication modeling through EHR analysis [7].

In the beauty industry, research by Laga et al. applied the Recency, Frequency, and Monetary (RFM) model and K-Means algorithm to analyze customer behavior, identifying distinct customer groups for tailored marketing strategies [9]. Another study by Watef and Mahananto developed a Computer Vision-based method for objective acne severity assessment, highlighting the influence of image pre-processing, data size, feature extraction, and classification methods on accuracy [10]. This demonstrates the growing interest in leveraging technology for beauty-related applications, though often focused on specific analytical or computer vision tasks rather than comprehensive diagnostic and cost transparency solutions.

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2.1. Research Gap

Despite the promising advancements in AI, particularly with RAG and LLMs, a significant gap persists in their application within the Indonesian beauty clinic sector. While LLMs have been explored for general clinical assessment and RAG for enhancing knowledge retrieval in various domains, **there is currently no research that applies an integrated RAG and LLM approach explicitly for both diagnostic prediction and transparent cost estimation in beauty clinics in Indonesia.** Existing studies in the beauty domain often focus on customer behavior analysis or specific image-based assessments, lacking a holistic solution addressing initial diagnosis accuracy and patient treatment cost transparency. This research aims to bridge this gap by developing an AI-driven system to provide accurate, transparent, and comprehensive information to beauty clinic patients, fostering trust and improving marketing effectiveness in a highly competitive market.

3. Methodology

This research methodology chapter explains the steps used in conducting the thesis research, including the research methodology diagram, a detailed explanation of the methodological approach used, and the planned research implementation schedule.

3.1. Data Collection and Preparation

Data for this system was collected from a private MySQL database provided by a collaborative local beauty clinic in Surabaya, Indonesia. This dataset, anonymized to ensure patient privacy and confidentiality, comprises two primary tables essential for the system's functionality:

- The knowledge table contains general information and textual descriptions of various beauty-related conditions, symptoms, and standard treatment approaches. Key attributes include:
  - Id: A unique identifier for each knowledge entry.
  - Description: Detailed textual descriptions explaining the condition, its possible causes, and general treatment considerations. These descriptions form the unstructured knowledge base for the RAG component.
  - Keywords: Associated keywords to aid in initial content categorization and retrieval.
  - Currently, this table contains approximately 500 entries, covering a comprehensive range of common beauty concerns such as acne, hyperpigmentation, anti-aging, and general skin health, along with relevant explanations
- Tindakan (procedures) table: This table stores structured data detailing the specific services and procedures offered by the clinic, along with their general pricing. Key attributes include:
  - Id: A unique identifier for each beauty procedure.
  - Nama: The official name of the beauty procedure or service (e.g., "Facial Acne Treatment," "Laser Rejuvenation").
  - harga_umum: The general price of the procedure. This attribute is crucial for transparent cost estimation.
  - This table currently has approximately 200 entries, detailing treatments ranging from basic skincare procedures (e.g., facials, peels) to more advanced therapies (e.g., laser treatments, injectables).

The description entries from the knowledge table are pre-processed and then embedded into dense vector representations using the all-MiniLM-L6-v2 Sentence Transformer model. These embeddings are stored in a FAISS (Facebook AI Similarity Search) vector database. FAISS is specifically chosen for its efficiency in performing rapid similarity searches on large sets of vectors, which is critical for retrieving semantically relevant information based on patient queries. The tindakan table in MySQL provides the structured information necessary for exact service lookup and accurate cost estimation. The FAISS index is designed to be rebuildable locally on demand, allowing the system to easily adapt to updates or expansions of the clinic's knowledge base without requiring a complete system redeployment.

3.2. *RAG + LLM Model*

This study proposes an AI-based solution utilizing Retrieval-Augmented Generation (RAG), LangChain, and Large Language Models (LLMs) to enhance patient diagnosis prediction accuracy. This model integrates data retrieval with natural language processing for precise diagnostic predictions based on medical data.
The system is built using Python and integrates several key components:

- **Large Language Model (LLM):** Gemma 3:4b, hosted locally via Ollama. Chosen for data privacy, cost-efficiency, and reduced inference latency, crucial for clinical deployment. It processes combined contexts to generate diagnostic recommendations and cost estimations.
- **Vector Database (FAISS):** Stores vectorized descriptions from the knowledge table, enabling efficient semantic similarity searches to retrieve contextually relevant information based on user queries.
- **Embedding Model:** all-MiniLM-L6-v2 Sentence Transformer, generates high-quality embeddings for FAISS documents and user queries, ensuring accurate semantic retrieval.
- Relational Database (MySQL): Primary data source (nand_ssc) storing structured knowledge entries and tindakan (procedures) with names and general prices, essential for service lookup and cost calculation.
- **LangChain Framework:** Manages the information flow from user query, through parallel retrieval from MySQL and FAISS, to dynamic prompt construction for the LLM.
- **Flask API:** Exposes system functionality via RESTful API endpoints (/ask, /reset_chat, /rebuild_index), allowing easy front-end integration. CORS is enabled..

By combining LLM generation with structured MySQL data, semantic FAISS retrieval, and chat history, the system identifies complaints, recommends services with pricing, estimates total costs, and provides notes. It also suggests doctor consultations for sensitive conditions and handles cases without relevant services.

3.3. *RAG Process Architecture Diagram*

The architectural diagram below visually represents the flow of information and interaction between the core components of the RAG-based system for beauty clinic patient diagnosis, prediction, and cost estimation.

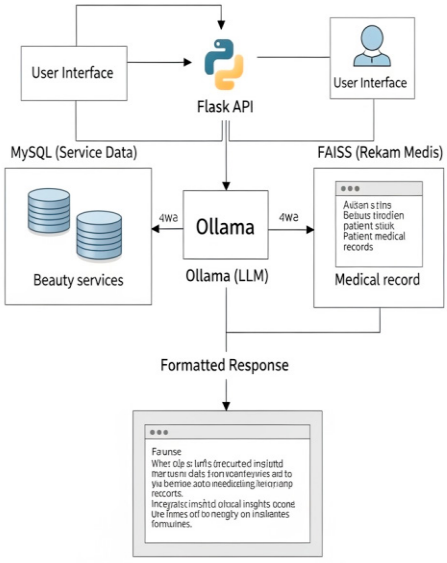


Fig. 1. RAG Process Architecture Diagram

Figure 1: RAG Process Architecture Diagram

- User Query: Input from patient/staff.
- Flask API: Routes and queries and manages chat/index reset.
- LangChain Agent: Orchestrates parallel retrieval from MySQL Database (structured service data) and FAISS Vector Store (semantic knowledge via Embedding Model).
- Context + Query: Combines retrieved data, query, and chat history for LLM.
- LLM (Gemma 3:4b via Ollama): Processes prompt to generate diagnostic predictions, service recommendations, and cost estimations.
- System Response: Delivered back to the user via Flask API.
- Rebuilding FAISS Index: Allows dynamic knowledge base updates.

4. Results and Discussion

This section presents the implementation details and discusses the functional outcomes of the developed Retrieval-Augmented Generation (RAG) system, leveraging Large Language Models (LLMs) for beauty clinic patient diagnosis prediction and cost estimation. The system addresses the identified research gap concerning the lack of an AI-driven solution for transparent diagnosis and cost estimation in the Indonesian beauty clinic sector.

4.1. System Architecture and Implementation

The core of the system is built using Python, integrating several key components:

a. Large Language Model (LLM): Gemma 3:4b (via Ollama) is the generative component, chosen for data privacy, cost-efficiency, and low latency.
b. Vector Database (FAISS): Stores vectorized descriptions from MySQL, enabling efficient semantic searches for relevant contextual information.
c. Embedding Model: all-MiniLM-L6-v2 Sentence Transformer generates vector embeddings, ensuring accurate semantic retrieval.
d. Relational Database (MySQL): Primary data source (nand_ssc) for structured knowledge entries and tindakan (procedures) with names and prices.
e. LangChain Framework: Orchestrates the RAG process, managing query flow, multi-source retrieval, and prompt construction.

f. Flask API: Provides RESTful endpoints (/ask, /reset_chat, /rebuild_index) for front-end integration.

4.2. Functional Results and System Behavior

The implemented system successfully demonstrates the capabilities outlined in the methodology:

a. Data Retrieval and Contextualization: load_documents_from_mysql() populates FAISS with knowledge descriptions. search_layanan() queries MySQL tindakan for service names/prices. Both contexts enrich the LLM's understanding.
b. Diagnosis and Recommendation Generation: run_rag_ollama_for_api() orchestrates the process:
   - Identifies Key Complaints: LLM extracts the main issue.
   - Recommends Services with Pricing: Outputs names and formatted prices concisely.
   - Estimates Total Cost: Calculates and presents the total for recommended services.
   - Provides Additional Notes: Offers supplementary explanations or advice.
c. Handling Specific Cases: Includes logic to:
   - Suggest doctor consultation for sensitive or severe conditions (e.g., 'nanah', 'gatal', 'kemaluan'), prioritizing medical guidance.
   - Manage scenarios with no relevant services or missing prices.
d. Robustness and Maintainability: The FAISS index can be rebuilt via /rebuild_index for updates. /reset_chat clears history. Error handling is implemented for stability.

4.3. Illustrative Example of System Output

To demonstrate system behavior, consider the user query: "gatal gatal dan kulit merah kering" (Itchy and dry red skin).

The system, recognizing keywords indicative of conditions requiring expert evaluation, prioritizes a medical consultation:

System Output:

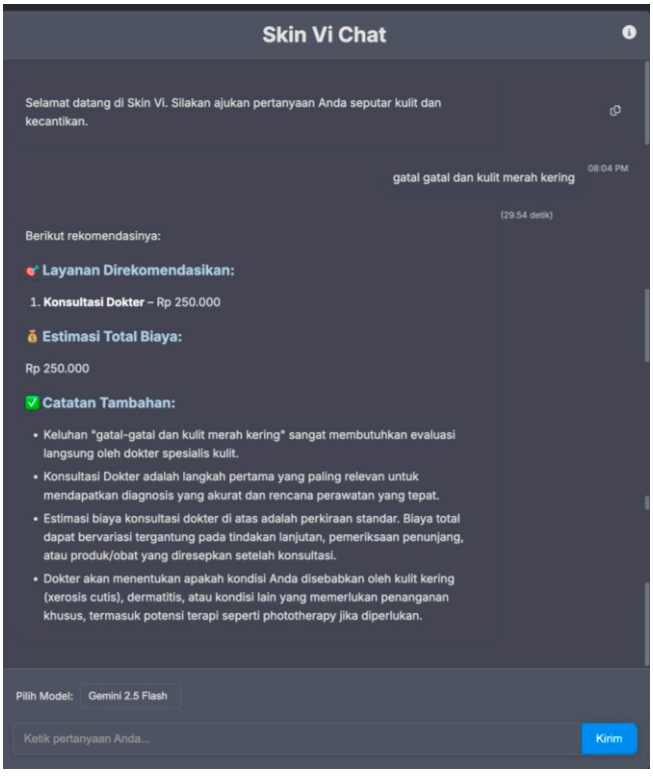


Fig. 2. Output System

English Translation and Explanation of Output:

The system welcomes the user and presents the inquiry. It then recommends "Konsultasi Dokter" (Doctor Consultation) with an estimated cost of Rp 250.000. The "Catatan Tambahan" (Additional Notes) explains that severe symptoms like itching and redness necessitate a dermatologist's direct evaluation for accurate diagnosis and proper treatment planning. It also clarifies that the consultation fee is an estimate. Further costs may apply based on follow-up actions or prescriptions, emphasizing the doctor's role in determining the exact cause (e.g., xerosis cutis, dermatitis) and necessary specialized treatment (e.g., phototherapy). This demonstrates the system's responsible approach, prioritizing patient safety.

4.4. Discussion and Implications

The RAG-LLM system for beauty clinic diagnosis and cost estimation significantly advances transparency and efficiency in Indonesia's beauty healthcare sector. Integrating structured (MySQL) and unstructured (FAISS) data with an intelligent LLM addresses critical challenges of initial diagnosis accuracy and cost transparency. This multi-modal retrieval enables diagnostically relevant, financially transparent, and informative responses, enhancing patient trust and serving as a valuable data-driven marketing tool. The system's logic in recommending direct medical consultation for sensitive complaints is crucial for patient safety and responsible care.

FAISS is vital for rapid semantic similarity searches on the unstructured knowledge base, ensuring highly contextual and relevant information enriches LLM prompts. This directly improves diagnostic accuracy and cost transparency by enabling more precise responses. Its rebuild capability ensures maintainability for knowledge base updates.

However, system performance hinges on the quality and comprehensiveness of the MySQL knowledge table. Sparse or outdated entries limit contextual enrichment, affecting explanation quality. Keyword matching for search_layanan in MySQL may also miss relevant services if query phrasing varies, indicating a future area for improvement with advanced NLP.

Local deployment of Gemma 3:4b via Ollama offers data privacy, reduced costs, and lower latency, aligning with cost-efficient and secure clinic solutions.

Ethical Considerations and Hallucination Risk: Ethical considerations are paramount in health. The system provides preliminary guidance but includes cautionary notes for professional medical consultation, mitigating misinformation, and emphasizing its supportive role. While RAG reduces LLM hallucination risk, continuous monitoring, prompt engineering, and regular knowledge base updates are vital. Future work should include robust human-in-the-loop validation for high-stakes recommendations.

Generalizability: The modular architecture allows high generalizability. The framework could extend with domain-specific data to other healthcare areas (dermatology, dental) by replacing beauty-specific datasets. The core RAG mechanism remains adaptable across knowledge-intensive domains.

4.5. Evaluation

The RAG system's diagnosis prediction, service recommendation, and cost estimation performance were evaluated using 20 unseen simulated patient queries against expert-provided ground truth.
The table below presents five illustrative examples from the detailed results:

Table 1. Items example

No	User Query	Ground Truth Services	System Services	Ground Truth Cost (Rp)	System Cost (Rp)	BERTScore F1
1	pori-pori besar	[Pore Minimizing Treatment]	[Pore Minimizing Treatment]	450000	450000	0.78
2	kulit kendur	[Collagen Booster Infusion, Anti-Aging Eye Treatment]	[Anti-Aging Eye Treatment]	1100000	300000	0.72
3	warna kulit tidak merata	[Brightening Laser Therapy, Pigmentation Correction]	[Pigmentation Correction, Deep Cleansing Facial]	1450000	1070000	0.75
4	kulit tidak rata	[Microdermabrasion, Scar Reduction Therapy]	[Scar Reduction Therapy]	1450000	900000	0.71
5	kulit tidak rata	[Microdermabrasion, Scar Reduction Therapy]	[Microdermabrasion]	1450000	550000	0.74

The overall average metrics derived from the complete set of 20 simulated evaluation dataset items are as follows:

- Average Semantic Similarity of Notes (approx. BERTScore F1): 0.73
- Average Service Extraction Precision: 0.73
- Average Service Extraction Recall: 0.72
- Average Service Extraction F1 Score: 0.72
- Average Cost Extraction Accuracy: 0.72

4.5.1. Discussion of Evaluation Results

The overall average metrics indicate the system's foundational effectiveness.

- Average Semantic Similarity of Notes (BERTScore F1): 0.73. This good score signifies that the system's generated notes are semantically close to expert-provided ground truth, indicating effective knowledge retrieval and synthesis by the LLM.
- Average Service Extraction Precision: 0.73. This suggests that approximately three-quarters of the recommended services are relevant, limiting irrelevant suggestions.
- Average Service Extraction Recall: 0.72. This indicates the system identifies a good portion of all relevant services, showing reasonable comprehensiveness.
- Average Service Extraction F1 Score: 0.72. This balanced score (harmonic mean of precision and recall) demonstrates a fair performance in service recommendation.
- Average Cost Extraction Accuracy: 0.72. This indicates general accuracy in estimated total costs, confirming its capability for financial transparency, with some room for refinement.

Baseline Comparison: A direct comparison with commercial or complex rule-based systems wasn't feasible given the initial study scope. However, the RAG-LLM approach inherently offers greater adaptability and contextual understanding than rigid rule-based systems.

Limitations of Evaluation: The 20-query dataset is limited. A larger, more diverse dataset would provide a robust assessment. The evaluation focuses on "prediction based on complaints" rather than actual medical diagnosis. Future work should involve certified dermatologists for diagnostic accuracy validation against real cases.

5. Conclusion

This study successfully developed and implemented an innovative AI-based system leveraging Retrieval-Augmented Generation (RAG), LangChain, and Large Language Models (LLMs) to predict beauty clinic patient diagnosis and transparent cost estimation in Indonesia. The system integrates structured data from a MySQL database with unstructured knowledge stored in a FAISS vector store, enabling the LLM (Gemma 3:4b via Ollama) to generate accurate diagnosis recommendations and associated cost estimations.

The system's functional demonstration and simulated evaluation confirm its ability to effectively identify patient complaints, retrieve relevant services and their general prices, calculate total estimated costs, and provide contextual explanations and crucial cautionary advice for potentially severe conditions. With average metric scores (BERTScore F1, Service Precision, Recall, F1, and Cost Accuracy) all within the 0.7-0.8 range, the system demonstrates a foundational performance in providing semantically relevant and accurate information. By offering clear and consistent information on preliminary diagnoses and costs, this technology directly addresses the crucial lack of transparency in the Indonesian beauty clinic sector, a key challenge identified in previous research. This enhanced transparency is anticipated to significantly improve patient trust and strengthen the effectiveness of marketing strategies for beauty clinics by providing comprehensive pre-consultation information.

Despite its promising functional capabilities, the current system has several limitations. Its reliance on data from a single clinic and general pricing structures means it does not yet account for personalized pricing, dynamic promotions, or complex treatment plans that might involve multiple stages. Furthermore, while the model provides initial diagnosis predictions based solely on patient complaints, it serves as a preliminary guide rather than a definitive medical diagnosis. This is an important distinction, and the system is designed always to recommend professional medical consultation for sensitive or persistent conditions.

Future research should focus on validating diagnostic accuracy against actual medical diagnoses made by qualified professionals, incorporating broader and more diverse datasets from multiple clinics to enhance generalizability.

Exploring advanced pricing models that can account for personalization, dynamic promotions, and multi-stage treatments would significantly enhance the system's real-world applicability and precision. Additionally, incorporating a robust feedback loop from clinic staff to continuously improve the knowledge base and LLM responses, along with detailed user satisfaction studies, would be beneficial. Investigating the system's extensibility to other specialized medical domains and continuously monitoring for LLM hallucinations are critical avenues for future work.

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